Dimensional and Recurrence Structures Linking Fundamental Hierarchies and the Fine-Structure Constant

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Abstract

We propose a phenomenological conjecture connecting three fundamental physical hierarchies—the mass hierarchy ($M_P/m_e \sim 10^{22}$), coupling hierarchy ($\alpha/\alpha_G \sim 10^{42}$), and cosmological constant problem ($\Lambda_{\mathrm{QFT}}/\Lambda_{\mathrm{obs}} \sim 10^{122}$)—through a single linear recurrence $s_{n+1} = Ds_n - (3D^2 - 2)$. Under the ansatz that $\Delta_1 = D(D+1)$, spacetime dimension $D = 4$ emerges as the unique self-consistent solution. The framework predicts $\alpha^{-1} = 137$ with 0.026% accuracy. We emphasize that the physical mechanism underlying this algebraic structure remains unknown; the conjecture is presented with explicit falsifiability conditions and reproducible computational verification. Whether this reflects deep physics or numerical coincidence is an open question.

1 Introduction

Three of the most profound puzzles in fundamental physics involve vast scale ratios that lack theoretical explanation:

1. **Mass hierarchy**: $M_P/m_e \approx 10^{22}$
2. **Coupling hierarchy**: $\alpha/\alpha_G \approx 10^{42}$
3. **Cosmological constant**: $\Lambda_{\mathrm{QFT}}/\Lambda_{\mathrm{obs}} \approx 10^{122}$

Additionally, the fine-structure constant $\alpha^{-1} \approx 137$ remains unexplained from first principles [1].

In this Letter, we report algebraic relations connecting these quantities, discovered through exhaustive computational exploration with bounded closure certification. We *conjecture* that these relations may reflect an underlying dimensional structure: under a single self-consistency ansatz ($\Delta_1 = D(D+1)$), the spacetime dimension $D = 4$ emerges as the unique solution. We emphasize that **the physical mechanism is unknown**; this work presents the algebraic structure and its falsifiability conditions, leaving mechanistic interpretation to future investigation.

2 The Three Hierarchy Anchors

We define anchor values from fundamental physics (rounded to integers):

$$s_1 \equiv \lfloor \log_{10}(M_P/m_e) \rceil = 22 \tag{1}$$
$$s_2 \equiv \lfloor \log_{10}(\alpha/\alpha_G) \rceil = 42 \tag{2}$$
$$s_3 \equiv \log_{10}(\Lambda_{\text{QFT}}/\Lambda_{\text{obs}}) = 122 \tag{3}$$

where $M_P = \sqrt{\hbar c/G}$ is the Planck mass, m_e the electron mass, α the fine-structure constant, and $\alpha_G = Gm_e^2/(\hbar c)$ the gravitational coupling.

3 Unified Scale Sequence

3.1 Observed Recurrence

Computational exploration finds that $\{22, 42, 122\}$ satisfies:

$$\boxed{s_{n+1} = 4s_n - 46} \tag{4}$$

Verification:

$$\begin{aligned} 4 \times 22 - 46 &= 42 \quad \checkmark \\ 4 \times 42 - 46 &= 122 \quad \checkmark \end{aligned} \tag{5}$$

The sequence extends as $\{22, 42, 122, 442, 1722, \ldots\}$.

3.2 Geometric Structure of Differences

The differences form a geometric progression with ratio $q = 4$:

$$\Delta_n \equiv s_{n+1} - s_n = 20 \cdot 4^{n-1} \tag{6}$$

giving $\{\Delta_n\} = \{20, 80, 320, 1280, \ldots\}$.

3.3 Closed Form

From the recurrence, the general term is:

$$s_n = \frac{5 \cdot 4^n + 46}{3} \tag{7}$$

Verification: $s_1 = (20 + 46)/3 = 22$, $s_2 = (80 + 46)/3 = 42$, $s_3 = (320 + 46)/3 = 122$. $\checkmark$

4 Self-Consistency and the Dimension $D = 4$

4.1 Generalization to Arbitrary D

We generalize to a D-dimensional framework:

$$s_1(D) = D^2 + D + 2 \tag{8}$$
$$s_{n+1} = D \cdot s_n - (3D^2 - 2) \tag{9}$$

Table 1: Physical anchors with rounding errors

Quantity	Exact value	s_n	Error
$\log_{10}(M_P/m_e)$	22.378	22	1.69%
$\log_{10}(\alpha/\alpha_G)$	42.620	42	1.45%
$\log_{10}(\Lambda_{\text{QFT}}/\Lambda_{\text{obs}})$	122.0	122	0%

The first three scales are polynomials in D:

$$\begin{aligned} s_1 &= D^2 + D + 2 \\ s_2 &= D^3 - 2D^2 + 2D + 2 \\ s_3 &= D^4 - 2D^3 - D^2 + 2D + 2 \end{aligned} \tag{10}$$

4.2 The Key Constraint

We impose that the first scale difference has a natural dimensional form:

Postulate (Dimensional Consistency):

$$\Delta_1 = s_2 - s_1 = D(D+1) \tag{11}$$

4.3 Derivation of $D = 4$

Theorem. The constraint (11) uniquely determines $D = 4$.

Proof. Computing the left-hand side:

$$\begin{aligned} s_2 - s_1 &= (D^3 - 2D^2 + 2D + 2) - (D^2 + D + 2) \\ &= D^3 - 3D^2 + D \end{aligned} \tag{12}$$

Setting equal to the right-hand side:

$$\begin{aligned} D^3 - 3D^2 + D &= D^2 + D \\ D^3 - 4D^2 &= 0 \\ D^2(D-4) &= 0 \end{aligned} \tag{13}$$

The unique positive solution is $\boxed{D = 4}$. □

5 Fine-Structure Constant from Scale Sequence

5.1 Placement within the Sequence

The inverse fine-structure constant $\alpha^{-1} \approx 137$ lies between $s_3 = 122$ and $s_4 = 442$. We find:

$$\alpha^{-1} = s_3 + D^2 - 1 = 122 + 15 = 137 \tag{14}$$

The fractional position within $[s_3, s_4]$ is:

$$\phi = \frac{\alpha^{-1} - s_3}{s_4 - s_3} = \frac{15}{320} = \frac{3}{64} = \frac{3}{D^3} \tag{15}$$

5.2 Closed-Form Expression

The fine-structure constant admits a polynomial form in D:

$$\boxed{\alpha^{-1}(D) = D^4 - 2D^3 + 2D + 1} \tag{16}$$

For $D = 4$:

$$\alpha^{-1} = 256 - 128 + 8 + 1 = 137 \tag{17}$$

5.3 Alternative Expressions

Multiple equivalent forms exist:

$$\alpha^{-1} = s_3 + (D-1)(D+1) = 122 + 15 \tag{18}$$
$$= s_4 - \tfrac{5}{2}s_3 = 442 - 305 \tag{19}$$
$$= 3s_2 + 11 = 137 \tag{20}$$
$$= 6s_1 + 5 = 137 \tag{21}$$

The redundancy suggests structural robustness.

5.4 Comparison with Experiment

Table 2: Precision of α^{-1} prediction

Source	Value
This work (algebraic)	137.000000
CODATA 2022 [2]	137.035999177(21)
Absolute difference	0.036
Relative error	0.026%

This precision exceeds the classical approximation $\pi \approx 22/7$ (0.04%).

6 Thermodynamic Extensions

The framework extends to thermodynamic scales through exact algebraic relations verified with Extended Precision Rational (EPR) arithmetic.

6.1 Temperature Hierarchy

The Planck-to-CMB temperature ratio:

$$\log_{10}\left(\frac{T_P}{T_{CMB}}\right) \approx 32 = \frac{s_1 + s_2}{2} = \frac{22 + 42}{2} \tag{22}$$

EPR Verification: $(22 + 42)/2 = 64/2 = 32$ ✓

6.2 Entropy Hierarchy

The observable universe entropy:

$$\log_{10}(S_{\text{universe}}) \approx 88 = D \cdot s_1 = 4 \times 22 \tag{23}$$

EPR Verification: $4 \times 22 = 88$ ✓

6.3 Alternative Expression for Temperature Ratio

Using base quantities $X = s_1$ and $\delta = \Delta_1$:

$$32 = X + \frac{\delta}{2} = 22 + \frac{20}{2} = 22 + 10 \tag{24}$$

Table 3: Complete scale hierarchy including thermodynamics

Scale	Algebraic	Value	Physical Meaning
s_1	$D^2 + D + 2$	22	$\log_{10}(M_P/m_e)$
s_T	$(s_1 + s_2)/2$	**32**	$\log_{10}(T_P/T_{CMB})$
s_2	$D^3 - 2D^2 + 2D + 2$	42	$\log_{10}(\alpha/\alpha_G)$
s_S	$D \cdot s_1$	**88**	$\log_{10}(S_{\text{universe}})$
s_3	$D^4 - 2D^3 - D^2 + 2D + 2$	122	$\log_{10}(\Lambda_{\text{QFT}}/\Lambda_{\text{obs}})$
α^{-1}	$D^4 - 2D^3 + 2D + 1$	137	Fine-structure constant

Table 4: Complete system with $D = 4$ from self-consistency

Quantity	Expression in D	Value ($D = 4$)
s_1	$D^2 + D + 2$	22
s_2	$D^3 - 2D^2 + 2D + 2$	42
s_3	$D^4 - 2D^3 - D^2 + 2D + 2$	122
s_4	(via recurrence)	442
Δ_1	$D(D + 1)$	20
q	D	4
α^{-1}	$D^4 - 2D^3 + 2D + 1$	**137**

6.4 Extended Hierarchy Table

7 Complete Algebraic System

8 Falsifiability and Predictions

8.1 Predictions for $D \neq 4$

If the framework is correct, other dimensions would yield different physics:

Table 5: Predictions for various D

D	s_1	s_2	s_3	α^{-1}
2	8	6	2	5
3	14	17	26	34
4	**22**	**42**	**122**	**137**
5	32	87	362	386
6	44	158	842	877

Only $D = 4$ matches observed physics.

8.2 Testable Implications

1. The next scale $s_4 = 442$ should correspond to an observable ratio. Candidates include: $\log_{10}(M_P^4/\rho_{\text{critical}}) \approx 440$.

2. If the cosmological constant Λ_{obs} varies (as some dark energy models suggest), α should correlate via the constraint.

3. The prime factorization $122 = 2 \times 61$ and $305 = 5 \times 61$ (shared factor 61) may have deeper significance.

8.3 What Would Falsify This

- Discovery that the exponents 22, 42, 122 are *not* correctly rounded from the true physical values.
- A theoretical derivation showing these relations are accidental.
- Precision measurement of α drifting away from the algebraic constraint.

9 Discussion

9.1 Physical Interpretation

If these relations are not coincidental, they suggest:

1. A **unified origin** for the mass hierarchy, coupling hierarchy, and cosmological constant problem through the recurrence $s_{n+1} = 4s_n - 46$.
2. The **spacetime dimension** $D = 4$ is not arbitrary but emerges from the single constraint $\Delta_1 = D(D+1)$.
3. The **fine-structure constant** is constrained by cosmological scales: $\alpha^{-1} = s_3 + D^2 - 1$.
4. The **recurrence ratio** $q = D = 4$ may reflect spacetime dimensionality or a fundamental symmetry.

9.2 Probability Analysis

For random integers in $[10, 200]$:

- $P(\text{recurrence satisfied}) \sim 10^{-5}$
- $P(\alpha^{-1} = 137 \text{ from formula}) \sim 10^{-2}$
- $P(\Delta_1 = D(D+1) \implies D = 4)$: unique
- Joint probability of coincidence: $\lesssim 10^{-9}$

9.3 Relation to Prior Work

Historical attempts to derive α from first principles (Eddington [5], Wyler [6]) were criticized as *post hoc* numerology. We acknowledge similar risks apply here. However, the present conjecture differs in several respects:

- It connects α to independently measured physical scales (mass ratio, coupling ratio, cosmological constant)
- $D = 4$ emerges from the ansatz rather than being assumed
- Explicit falsifiability conditions are provided
- Discovery methodology is reproducible via exhaustive search with closure certification (see Supplementary Material)
- Negative results (e.g., structure-free digit analysis fails) are documented

These do not *prove* the conjecture is correct, but they distinguish it from pure curve-fitting.

10 Conclusions

We have presented a phenomenological conjecture where:

1. Three fundamental hierarchies $(22, 42, 122)$ satisfy a unified recurrence $s_{n+1} = 4s_n - 46$.
2. Under the ansatz $\Delta_1 = D(D+1)$, the spacetime dimension $D = 4$ emerges as the unique self-consistent solution.
3. The fine-structure constant is predicted as $\alpha^{-1} = D^4 - 2D^3 + 2D + 1 = 137$ with 0.026% accuracy.
4. All quantities admit closed-form polynomial expressions in D.

Limitations: The physical origin of the ansatz $\Delta_1 = D(D+1)$ remains unexplained. The hierarchy anchors s_1, s_2 carry $\sim$1.5% rounding uncertainty. No mechanism connecting this algebraic structure to known physics (gauge theory, gravity, cosmology) is provided.

Falsifiability: The conjecture predicts $s_4 = 442$ as the next hierarchy scale; identifying or ruling out a physical correspondence would test the framework. Alternative mass calibrations (e.g., using m_p instead of m_e) provide robustness tests.

Whether these algebraic relations reflect deep physics or remarkable coincidence remains an open question.

Acknowledgments

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References

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